

Ultrahigh-sensitive anomalous Hall effect to magnetic field enabled by an irradiation-induced defect engineering

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Yu Qi,¹ Boyi Wang,¹ Fei Meng,¹ Xutong Meng,¹ Lei Ding,² Xuewei He,¹ Yi Cao,¹ Yang Liu,³ Youqiang Xu,⁴ Chun Feng,^{1,a)} Xingzhong Cao,^{5,a)} and Guanghua Yu¹

AFFILIATIONS

¹School of Materials Science and Engineering, University of Science and Technology Beijing, Beijing 100083, China

²School of Materials Science and Engineering, Hainan University, Haikou, Hainan 570228, China

³School of Chemistry and Biological Engineering, University of Science and Technology Beijing, Beijing 100083, China

⁴Key Laboratory of Geriatric Nutrition and Health (Beijing Technology and Business University), Ministry of Education, Beijing 100048, China

⁵Institute of High Energy Physics, Chinese Academy of Sciences, Beijing 100049, China

^{a)}Authors to whom correspondence should be addressed: fengchun@ustb.edu.cn and caoxzh@ihep.ac.cn

ABSTRACT

Ultrahigh-sensitivity linear magnetic sensors that utilize the anomalous Hall effect (AHE) are crucial for weak magnetic field sensing, quantum computing, biological detection, and spintronic devices. Presently, the AHE tunability primarily relies on accurate composition control during the film deposition, which typically limits the magnetic field sensitivity to below 35 kΩ/T. Here, we propose a promising strategy for boosting the AHE by artificially introducing crystal defects through γ -ray irradiation. By subjecting the Ta/CoFe/MgO/Ta multilayer to γ -ray irradiation with an irradiation at a dose of 500 Gy, an appropriate concentration of point defects can be introduced not only to increase electron scattering and strengthen both the skew scattering and side-jump contributions but also to modify the Berry curvature to enhance the intrinsic contribution, leading to a substantial AHE improvement. Meanwhile, these point defects also promote interfacial atomic diffusion, inducing enhanced Fe–O hybridization and orbital reconstruction that reduce the out-of-plane saturation field. As a result, a linear sensor with an ultrahigh sensitivity of 47.4 kΩ/T was achieved. These findings elucidate the defect–transport coupling mechanism and offer a new paradigm for modulating spin-dependent transport through defect engineering.

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Linear sensors based on the anomalous Hall effect (AHE) offer the characteristics of high sensitivity, rapid response, and strong anti-interference capability, making them promising for magnetic field sensing, quantum computing, biological detection, and spintronic devices.^{1–3} For weak-field detection in micro- and nanoscale devices (e.g., the field of a single electron is about 5 fT,⁴ and the biomagnetic field of a human ranges from pT to fT levels,^{5,6}) AHE linear sensors require both a large anomalous Hall resistance (R_{AH}) and a small saturation field. The AHE arises from intrinsic contributions (related to Berry curvature) and extrinsic contributions (from impurity scattering).⁷ The intrinsic contribution is determined by the inherent energy band structure of the material. For example, the enhancement of spin–orbit coupling (SOC) by altering composition can reconstruct the electronic structure and modify the Berry curvature, thereby

increasing the R_{AH} .^{8–11} The extrinsic contribution stems mainly from electron scattering at grain boundaries, impurities, and interfaces. Inserting nonmagnetic spacer layers or increasing the period number of multilayer film modifies the scattering potential and adds interfacial scattering centers that raise the probability of asymmetric scattering and increase the R_{AH} .^{12–16} The magnetic field sensitivity also depends on the magnetic anisotropy, which can be tuned by introducing heavy metals to enhance SOC,¹⁷ inserting oxides to induce interfacial orbital hybridization,^{18–20} applying electric fields to regulate interfacial charge distribution,^{21,22} or applying stress or nitrogen doping to alter the coordination environment.^{23,24} Such tuning lowers the field required to magnetize the sensor and thus improves the sensitivity. Overall, AHE modulation currently relies on accurate composition control during the film deposition, which typically limits the

sensitivity below $35 \text{ k}\Omega/\text{T}$.^{9,10,12–19,25–29} Therefore, it is desirable to further enhance the sensitivity to meet the requirement of high-precision detection of weak magnetic fields.

Crystal defects can significantly influence the electronic structure and electron transport of materials, effectively modulating their magnetic, electrical, catalytic, and optical properties.^{30–34} Therefore, the introduction of defects offer a promising route to modulate the AHE for constructing highly sensitive magnetic sensors. However, naturally occurring defects during the film deposition are typically random and poorly controlled spatially, limiting both the efficacy and stability of AHE modulation. In this study, we substantially enhance the AHE by artificially introducing the crystal defects via γ -ray irradiation. Subjecting Ta/CoFe/MgO/Ta multilayers to an appropriate γ -ray irradiation introduces a well-controlled concentration of point defects, which increases electron scattering to strengthen both the skew scattering and side-jump contributions and modifies the Berry curvature to enhance the intrinsic contribution, remarkably improving the AHE. Moreover, these defects also promote interfacial atomic diffusion, inducing the enhanced Fe–O hybridization and orbital reconstruction that reduce the out-of-plane saturation field. Ultimately, a linear magnetic sensor with an ultrahigh sensitivity of $47.4 \text{ k}\Omega/\text{T}$ is realized.

Ta(0.83 nm)/Co₂₀Fe₈₀(0.7 nm)/MgO(0.8 nm)/Ta(1 nm) multilayer film with small coercivity and tunable magnetic anisotropy^{35,36} was deposited on Si/SiO₂ substrate using magnetron sputtering at room temperature. The base pressure in the sputtering chamber was maintained below 3×10^{-7} mTorr, and the pressure of Ar during

sputtering was 4 mTorr. After deposition, the sample was annealed at 100°C for 30 min in a vacuum furnace with a base pressure lower than 5×10^{-5} Pa. Then, the annealed sample was subjected to γ -ray irradiation from a ^{60}Co source at cumulative doses (D) of 500, 1000, and 1500 Gy. Finally, the sample was patterned into a Hall bar structure with a length of $46 \mu\text{m}$ and a width of $10 \mu\text{m}$ [Fig. 1(a)]. To study the effect of γ -ray irradiation on AHE, the Hall response curves [Figs. 1(b)–1(e)] were measured by applying a constant current of 1 mA along the long axis of the Hall bar and an external magnetic field varying between ± 40 Oe perpendicular to the film plane. All curves exhibit good linearity in the low-field range (± 5 Oe), which is beneficial for constructing a highly sensitive linear sensor. The magnetic field sensitivity (S_v) is obtained from the slope of the linear region using the formula $S_v = dR_{\text{AH}}/dH$,³⁷ where R_{AH} is the anomalous Hall resistance and H is the applied magnetic field. Figure 1(f) shows the variation of S_v with D . Clearly, the S_v first rises and then falls as D increases. Specifically, S_v of the unirradiated sample ($D = 0$ Gy) is $38.3 \text{ k}\Omega/\text{T}$, which was optimized by tuning the CoFe thickness and annealing temperature (Fig. S1 in the supplementary material). S_v increases to $47.4 \text{ k}\Omega/\text{T}$ after a moderate irradiation dose of 500 Gy but reduces to $35.3 \text{ k}\Omega/\text{T}$ at 1500 Gy that is comparable to the unirradiated sample. Furthermore, the device-to-device and batch-to-batch statistical data were measured to exclude the influence of sample inhomogeneity on the S_v (Figs. S2 and S3 in the supplementary material). Therefore, an appropriate irradiation effectively enhances the AHE. Notably, the maximum sensitivity of $47.4 \text{ k}\Omega/\text{T}$ surpasses that of AHE sensors reported over the past 15 years, as shown in

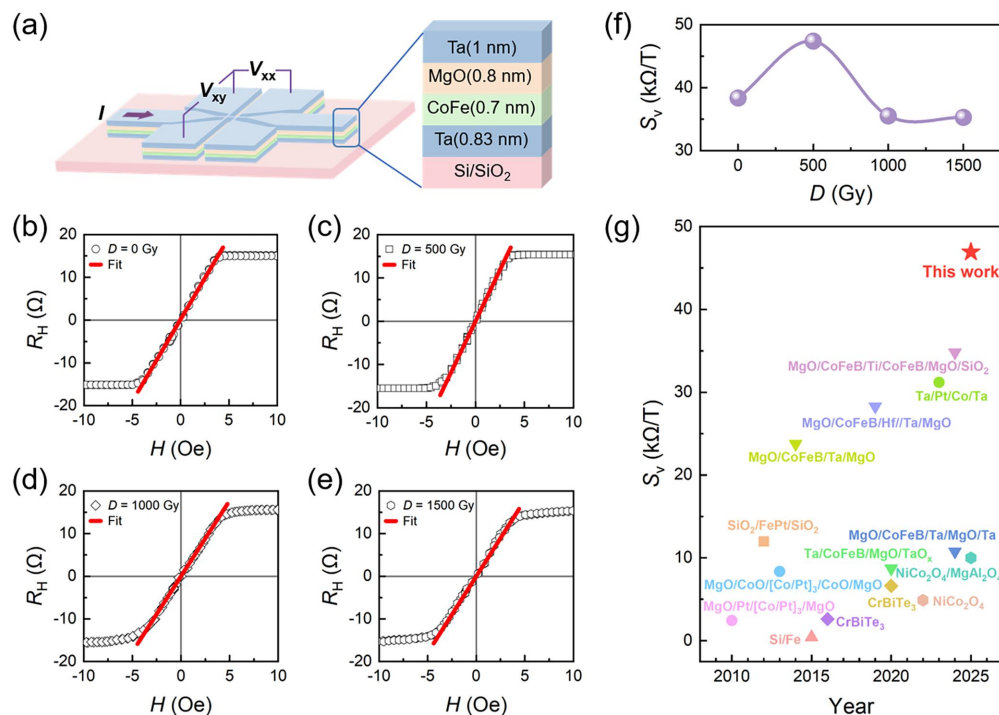


FIG. 1. Effect of γ -ray irradiation on AHE. (a) Schematic of the thin-film structure and AHE curve measurement. (b)–(e) AHE curves of the Ta(0.83 nm)/Co₂₀Fe₈₀(0.7 nm)/MgO(0.8 nm)/Ta(1 nm) sample after the irradiation at different D : (b) $D = 0$ Gy; (c) $D = 500$ Gy; (d) $D = 1000$ Gy; (e) $D = 1500$ Gy. (f) Variation of S_v with D . (g) Comparison of sensor sensitivity over the past 15 years.

Fig. 1(g).^{9,10,12–19,25–29} In addition, we also observed that the linear slope beyond the saturation in Figs. 1(b)–1(e) varies with irradiation dose. The slope reflects the ordinary Hall coefficient $R_0 = -1/(ne)$, where n is the carrier concentration, and e represents the electric charge of a single electron.³⁸ The calculated n value rises from $3.4 \times 10^{20} \text{ cm}^{-3}$ in the unirradiated sample to $4.9 \times 10^{20} \text{ cm}^{-3}$ at $D = 500 \text{ Gy}$, and then decreases to $3.0 \times 10^{20} \text{ cm}^{-3}$ at $D = 1500 \text{ Gy}$. This non-monotonic evolution results in the reduction of the post-saturation linear slope at low doses and the subsequent enhancement at high doses. Since the ordinary Hall resistance is much smaller than the anomalous Hall resistance in ferromagnets, we will focus on the underlying mechanism of the irradiation effect on the AHE and S_v in detail.

To clarify how γ -irradiation affects the defect concentration in the film, positron annihilation spectroscopy (PAS) was performed on the $\text{Ta}(0.83 \text{ nm})/[\text{CoFe}(0.7 \text{ nm})/\text{MgO}(0.8 \text{ nm})]_{15}/\text{Ta}(1 \text{ nm})$ sample using a ^{22}Na radioactive source as the positron source to generate the incident positrons with a continuously tunable incident energy. The incident high-energy positrons (e^+) are captured and annihilated by electrons (e^-) near defects, emitting γ -photons (499.5–522.5 keV) that generate the Doppler broadening spectrum [Fig. 2(a)].³⁹ The annihilation properties are characterized by the S and W parameters derived from this spectrum. The S parameter—the count ratio within 510.2–511.8 keV relative to the total peak (499.5–522.5 keV)—corresponds to the annihilations with low-momentum electrons associated with defects. The W parameter—the count ratio within 513.6–516.9 and 505.1–508.4 keV relative to the total peak—represents the annihilations with high-momentum electrons associated with the perfect

lattice.^{40–42} Figures 2(b) and 2(c) show the dependences of S and W parameters on the incident positron energy for different irradiation doses. Signals in the 0.6–1.6 keV range mainly derive from the $[\text{CoFe}/\text{MgO}]_{15}$ multilayer. Here, S increases with D , while W exhibits the opposite trend, indicating that the defect concentration in the film rises with irradiation dose. The linear W - S relationship in Fig. 2(d) further reveals the defect types and positron annihilation sites.⁴³ Clearly, the W - S plots for all samples are well fitted by two straight lines, implying two distinct annihilation sites linked to point defects such as vacancies or interstitials.⁴⁴ The similar curve shapes indicate consistent defect types with increasing dose, while their dose-dependent shift toward higher S and lower W may be due to the progressive clustering of isolated vacancies into vacancy clusters with increasing dose.⁴³

Having established how defects evolve during irradiation, we next study how they affect the AHE. To identify the contributing mechanisms, longitudinal and Hall resistances were measured at various temperatures. Figures 3(a) and 3(b) illustrate the temperature (T) dependence of the longitudinal resistivity (ρ_{xx}) and anomalous Hall resistivity (ρ_{AH}) for the different doses. As T increases, ρ_{xx} decreases monotonically for all samples, while ρ_{xx} initially rises and then falls with increasing dose. Likewise, ρ_{AH} decreases monotonically with T and increases slightly with increasing dose. In ferromagnetic materials, the normalized anomalous Hall resistance $R_s = \frac{\rho_{AH}}{4\pi M_s} = a\rho_{xx} + b\rho_{xx}^2$ was commonly used to relate the anomalous Hall resistivity to the longitudinal resistivity,⁴⁵ where M_s is the saturation magnetization. The first-order coefficient a represents the skew scattering contribution, and the second-order coefficient b represents the combined intrinsic and

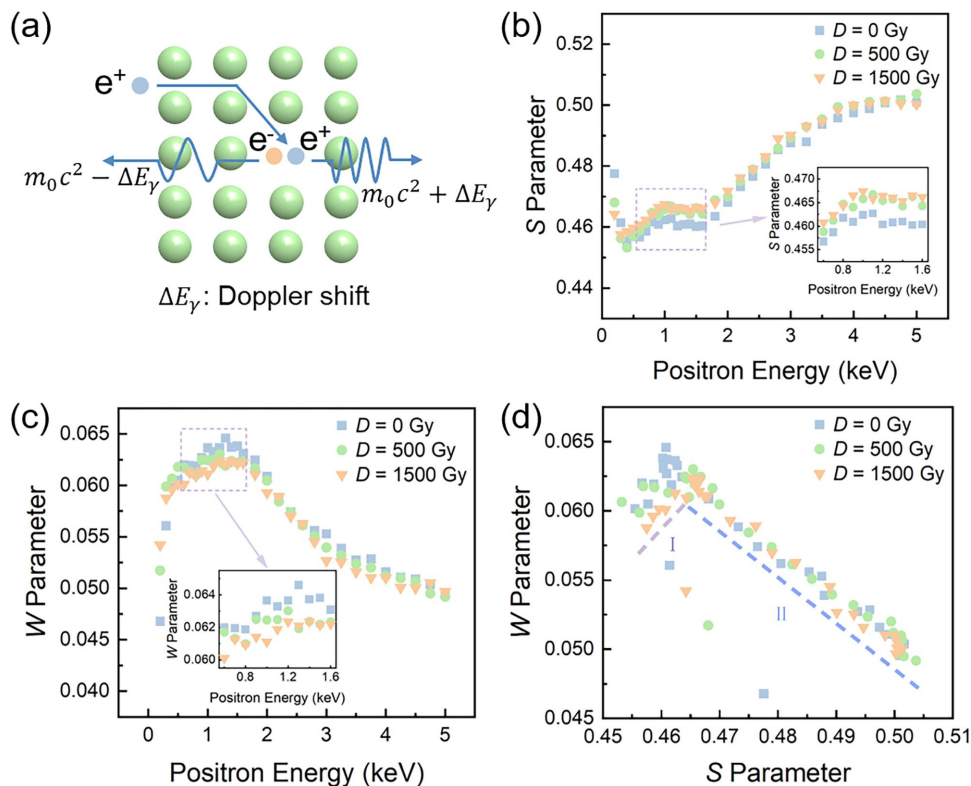


FIG. 2. Evolution of defects in the thin film during irradiation. (a) Schematic of the principle for positron annihilation test. (b) Variation of the S parameter with incident positron energy. Inset: an enlarged view of S variation in the 0.6–1.6 keV range. (c) Variation of the W parameter with incident positron energy. Inset: an enlarged view of W variation in the 0.6–1.6 keV range. (d) Dependence of the W parameter on the S parameter.

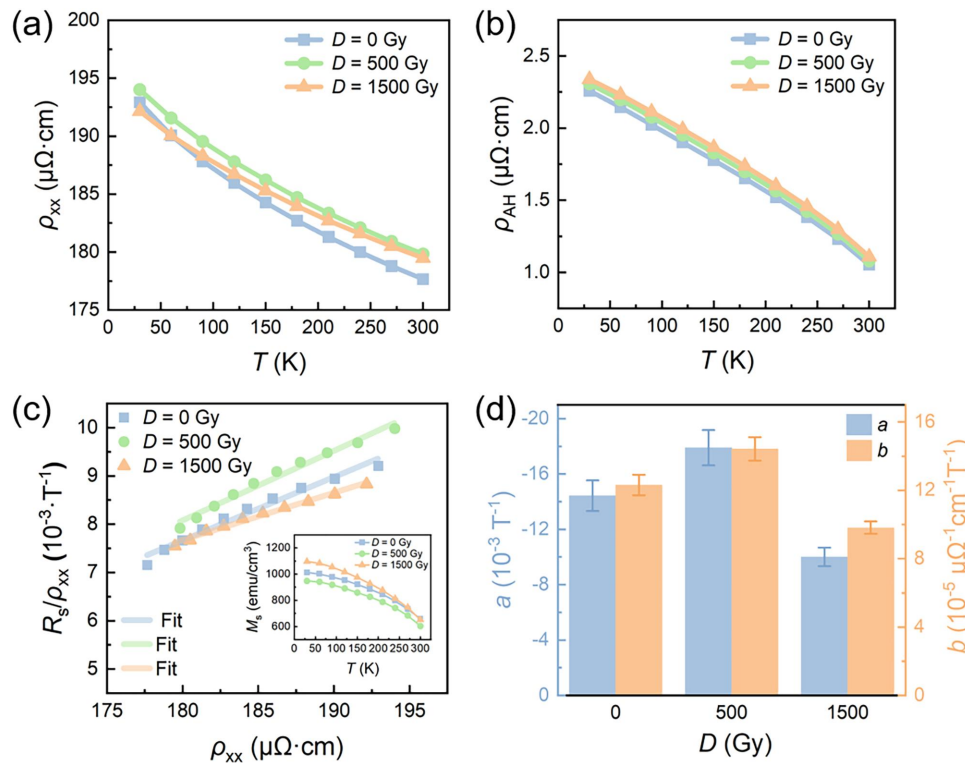


FIG. 3. Influence of defects on the AHE. (a) Variation of ρ_{xx} with temperature T . (b) Dependence of ρ_{AH} on T . (c) Variation of R_s/ρ_{xx} with ρ_{xx} . Inset: variation of M_s with T . (d) Dependence of coefficients a and b with D .

side-jump contributions. To determine a and b , the $R_s/\rho_{xx}-\rho_{xx}$ curve is plotted in Fig. 3(c), and the coefficients calculated through linear fitting are displayed in Fig. 3(d).

For the unirradiated sample, the coefficients a and b are $-14.44 \times 10^{-3} \text{ T}^{-1}$ and $12.31 \times 10^{-5} \mu\Omega^{-1}\text{cm}^{-1}\text{T}^{-1}$, respectively. Upon γ -ray irradiation at 500 Gy, a decreases to $-17.91 \times 10^{-3} \text{ T}^{-1}$, and b rises to $14.43 \times 10^{-5} \mu\Omega^{-1}\text{cm}^{-1}\text{T}^{-1}$. This result reflects that the introduction of optimal point defects can act as effective electron-scattering centers and enhance the scattering probability, leading to the increment of skew scattering contribution and raising of $|a|$. Meanwhile, these defects distort the lattice symmetry and strengthen the local defect potential fields, increasing the probability of electron side-jump scattering and modifying the Berry curvature, leading to the elevation of b . Therefore, a marked enhancement in R_{AH} is observed. When the D further increases to 1500 Gy, a increases sharply to $-10.01 \times 10^{-3} \text{ T}^{-1}$ and b reduced significantly to $9.82 \times 10^{-5} \mu\Omega^{-1}\text{cm}^{-1}\text{T}^{-1}$. This observation indicates that the excessive irradiation at 1500 Gy may lead the individual vacancies to aggregate into vacancy clusters, decreasing the electron scattering and diminishing both the skew scattering contribution and $|a|$. On the other hand, the introduction of point defects may lead to a reduction in both the Berry curvature and the gradient of the defect potential field. The combined effect of these two factors weakens the contributions of the intrinsic mechanism and the side-jump mechanism, thereby reducing b . On this basis, a pronounced reduction of R_{AH} is observed. In summary, the introduction of appropriate point defects

during the γ -ray irradiation can not only increase the electron scattering and strengthen both the skew scattering and side-jump contributions but also modify the Berry curvature to enhance the intrinsic contribution, which are beneficial for enhancing the AHE.

Moreover, the magnetic field sensitivity is also associated with the magnetic field for out-of-plane magnetization, which is mainly determined by the magnetic anisotropy and coercivity. Hysteresis loops of the films [Figs. 4(a)–4(d)] were measured at room temperature using a physical property measurement system (PPMS) by applying the in-plane or out-of-plane fields within ± 8 kOe. The anisotropy was quantified by the effective magnetic anisotropy constant (K_{eff}), obtained from the area between the in-plane and out-of-plane magnetization curves. $K_{\text{eff}} > 0$ indicates perpendicular magnetic anisotropy (PMA). Figure 4(e) shows the dependence of K_{eff} on D . The K_{eff} of the unirradiated sample is $0.82 \times 10^6 \text{ erg/cm}^3$, which rises markedly to $1.18 \times 10^6 \text{ erg/cm}^3$ at 500 Gy, then decreases to $1.08 \times 10^6 \text{ erg/cm}^3$ at 1500 Gy. Thus, the irradiation causes an obvious effect on the magnetic anisotropy, in which the strongest PMA with the largest K_{eff} occurs when the sample is irradiated at 500 Gy. The influence of magnetic anisotropy on the out-of-plane magnetization field will be discussed in detail later. Moreover, the coercivity can also affect the magnetization field. In general, the irradiation-introduced defects function as pinning sites during the magnetization process, enhancing coercivity and hindering domain motion.^{46,47} The variation of in-plane and out-of-plane H_c with the irradiation dose is shown in Fig. 4(f). The out-of-plane H_c values remain within 1 Oe and do not

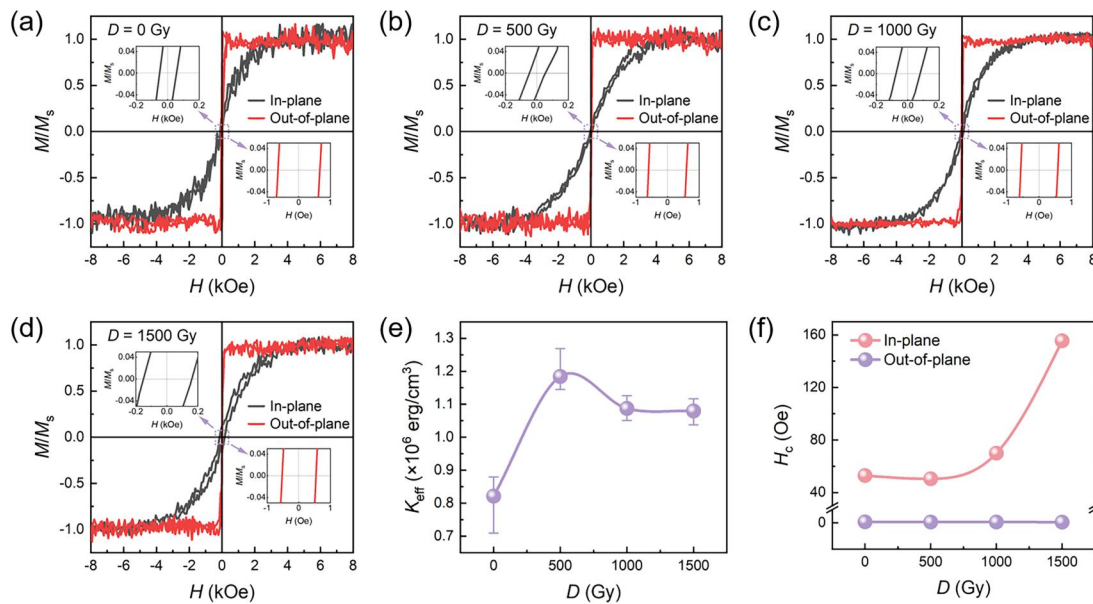


FIG. 4. Effect of irradiation on magnetic anisotropy. (a)–(d) Normalized M - H curves of the films treated at different irradiation doses: (a) $D = 0$ Gy; (b) $D = 500$ Gy; (c) $D = 1000$ Gy; (d) $D = 1500$ Gy. Insets: enlarged views of the in-plane M - H curves in the range of -0.2 to 0.2 kOe and the out-of-plane M - H curves in the range of -1 to 1 Oe. (e) Variation of K_{eff} with D . (f) Dependence of H_c on D .

show a significant change at present irradiation doses. Instead, the in-plane H_c increases, indicating that the irradiation mainly affects the in-plane coercivity instead of out-of-plane coercivity. Since the S_v is mainly determined by the out-of-plane magnetization, the defect-induced domain pinning is not the main reason for the S_v tunability.

To reveal the origin of the magnetic anisotropy variation, x-ray photoelectron spectroscopy (XPS) was used to characterize the changes in the chemical states before and after γ -ray irradiation. An Al $K\alpha$ monochromatic x-ray source was used, and the passing energy of the energy analyzer was 100 eV. Prior to analysis, all samples underwent Ar^+ etching for 20 s at an approximate rate of 0.05 nm/s to remove surface contaminants. Because the Co content is much lower than that of the Fe content, we focus on the chemical state change of Fe. Figure 5(a) presents the Fe 2p high-resolution spectra at the CoFe/MgO interface. By spectral fitting, the metallic Fe, FeO_x ($0 < x < 1$), and FeO peaks can be identified at 706.8, 707.8, and 709.9 eV, respectively. Furthermore, the proportions of metallic and oxidized Fe are obtained using the area ratio of fitted peaks [Fig. 5(b)]. For the unirradiated sample, metallic and oxidized Fe (FeO_x and FeO) account for 52.36% and 47.64%, indicating a substantial Fe–O hybridization in the as-deposited film. Moderate irradiation ($D = 500$ Gy) raised the fraction of oxidized Fe to 52.83%, signaling the enhanced Fe–O hybridization. However, the excessive γ -irradiation at 1500 Gy lowers the fraction to 48.98%, indicating the weakened Fe–O hybridization. These findings can be verified by x-ray absorption spectroscopy (XAS) measurements, which were performed in total electron yield (TEY) mode on the BL08U1A beamline at the Shanghai Synchrotron Radiation Facility (SSRF) with a vacuum better than 5×10^{-8} Pa, a soft x-ray photon flux of 1.5×10^8 photons/s, and an energy resolution ($E/\Delta E$) better than 17 000. Figure 5(c) presents the normalized Fe L -edge XAS spectra before and after irradiation.

The characteristic L_3 and L_2 peaks near 707.9 and 720.5 eV are observed.⁴⁸ The intensity ratio of L_3 to L_2 peaks reflects the degree of Fe–O hybridization. Generally, a larger ratio indicates the stronger Fe–O hybridization.⁴⁹ The L_3/L_2 ratio increases when D increases to 500 Gy, and then decreases with enhancing D to 1500 Gy, confirming that the strongest Fe–O hybridization emerges at 500 Gy. Clearly, the γ -irradiation effectively modulates Fe–O hybridization at the CoFe/MgO interface.

Furthermore, x-ray linear dichroism (XLD) can be obtained from the difference between XAS spectra measured with the light polarization vector parallel ($E//a$) and perpendicular ($E//c$) to the sample surface, enabling us to study the orbital occupancy of Fe 3d electrons. The integrated area (A_{XLD}) of the XLD spectrum under the L_2 peak (718–726 eV, marked by the dashed box) was utilized to evaluate the orbital occupancy of Fe 3d electrons. Generally, a positive A_{XLD} indicates that Fe 3d electrons preferentially occupy the out-of-plane symmetric d_{z^2} orbital, whereas a negative A_{XLD} indicates the preferential occupation of the in-plane symmetric $d_{x^2-y^2}$ orbital.^{22,50–52} Figure 5(d) shows the Fe XLD spectra before and after irradiation. Figure 5(e) summarizes the variation of A_{XLD} with D . For the unirradiated sample, A_{XLD} is -0.02 , indicating preferential occupation of the in-plane $d_{x^2-y^2}$ orbital. After γ -ray irradiation at 500 Gy, A_{XLD} increases to 0.07, reflecting the transformation of orbital occupancy from the in-plane $d_{x^2-y^2}$ to out-of-plane d_{z^2} . With further increment of D to 1500 Gy, A_{XLD} decreases to 0.02, indicating that excessive irradiation transfers electrons from the d_{z^2} orbital back to the $d_{x^2-y^2}$ orbital. These findings demonstrate that the γ -ray irradiation induces significant orbital reconstruction by modulating Fe–O hybridization at the CoFe/MgO interface.

Combining the above results, we can explain the physical origin of magnetic anisotropy evolution and its influence on the linear

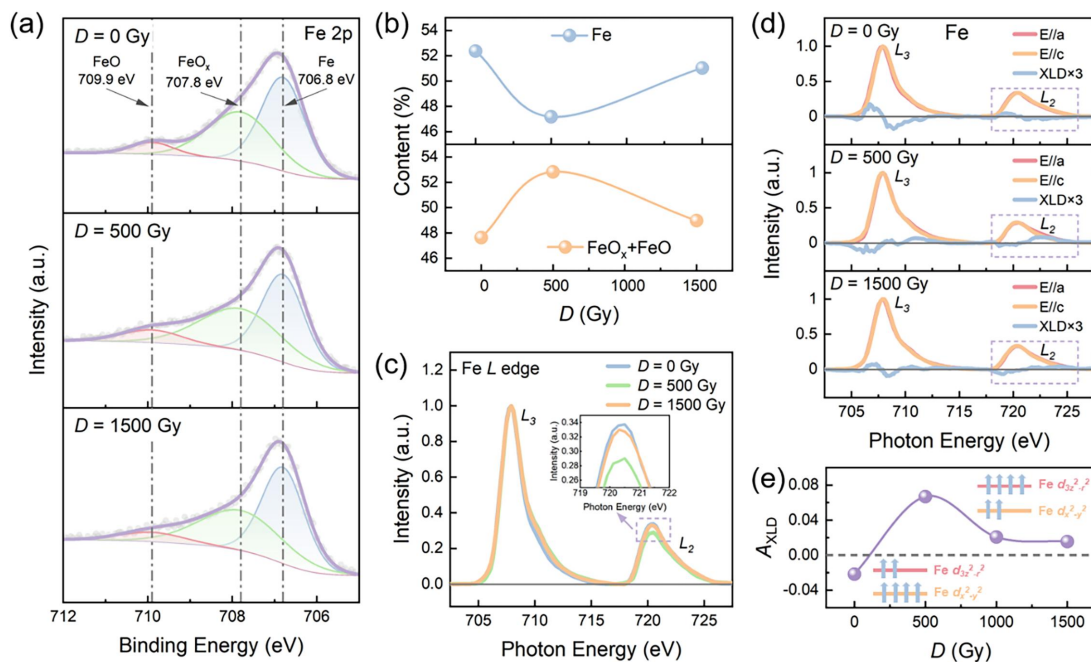


FIG. 5. Variation of interfacial orbital hybridization and orbital occupancy. (a) Fe 2p XPS spectra at the CoFe/MgO interface. (b) Variation of the proportions of metallic Fe and oxidized Fe with D . (c) Normalized Fe L-edge XAS spectra at the interface. Inset: an enlarged view of the XAS spectra in the range of 719–722 eV. (d) Normalized Fe L-edge XLD spectra at the interface. The dashed box indicates the energy range used to calculate the integrated area A_{XLD} . (e) Variation of A_{XLD} with D . Inset: schematic diagram of d-orbital occupancy for different A_{XLD} .

sensitivity of the AHE sensor. The optimal irradiation ($D = 500$ Gy) promotes interdiffusion of interfacial O and Fe atoms, substantially enhancing Fe–O hybridization and increasing the out-of-plane orbital occupancy.⁵³ Both theoretical and experimental results demonstrate that optimized out-of-plane orbital occupancy of Fe is beneficial for enhancing PMA.^{24,54,55} On the other hand, Compton scattering during irradiation transfers energy between photons and electrons, intensifying the interaction of high-energy free spin electrons with the lattice and further enhancing PMA.^{46,56} The strengthened PMA markedly reduces the magnetic field for out-of-plane magnetization. Meanwhile, the point defects obviously increase the electron scattering and boost the AHE (we discussed before). Finally, the linear sensitivity of the AHE sensor is improved significantly. However, under excessive irradiation (1000 or 1500 Gy), the aggregation of vacancies into vacancy clusters reduces electron scattering and weakens the AHE. Meanwhile, secondary electrons from Compton scattering are proposed to lower the PMA by disrupting the optimal Fe–O hybridization through electron transfer.^{57–59} Thus, the linear sensitivity is reduced. In summary, appropriate point defects enhance the AHE by increasing electron scattering and lower the out-of-plane saturation field by modulating interfacial Fe–O hybridization, significantly improving the magnetic field sensitivity of the linear AHE sensor.

In this work, γ -ray irradiation was used to introduce controllable point defects into the Ta/CoFe/MgO/Ta film, substantially enhancing both the AHE and linear sensitivity. Appropriate point defects not only increase the electron scattering and strengthen both the skew

scattering and side-jump contributions but also modify the Berry curvature to enhance the intrinsic contribution, markedly improving the AHE. Meanwhile, the defects also promote interfacial atomic diffusion, enhancing Fe–O hybridization and orbital reconstruction and thereby reducing the out-of-plane saturation field. As a result, an AHE linear sensor with an ultrahigh sensitivity of 47.4 k Ω /T was achieved. These findings elucidate the defect–transport coupling mechanism and provide a new paradigm for modulating spin-dependent transport properties through defect engineering.

See the [supplementary material](#) for the following: Transport and magnetic anisotropy were previously optimized by tuning CoFe thickness (t_{CoFe}) and annealing temperature (T) (Fig. S1). Statistical data across devices and batches serve to demonstrate sample uniformity (Figs. S2 and S3).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yu Qi: Conceptualization (lead); Data curation (lead); Investigation (lead); Methodology (lead); Visualization (lead); Writing – original draft (lead). **Boyi Wang:** Data curation (supporting); Investigation (supporting); Methodology (supporting). **Fei Meng:** Data curation (supporting); Methodology (supporting). **Xutong Meng:** Investigation (supporting); Methodology (supporting). **Lei Ding:** Investigation (supporting); Validation (supporting). **Xuewei He:** Investigation (supporting); Methodology (supporting). **Yi Cao:** Methodology (supporting); Resources (supporting); Validation (supporting). **Yang Liu:** Resources (supporting); Validation (supporting). **Youqiang Xu:** Methodology (supporting); Validation (supporting). **Chun Feng:** Conceptualization (lead); Funding acquisition (lead); Methodology (lead); Project administration (lead); Resources (lead); Supervision (lead); Validation (lead); Writing – review & editing (lead). **Xingzhong Cao:** Methodology (supporting); Resources (supporting). **Guanghua Yu:** Conceptualization (supporting); Supervision (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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